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# Artem Makoian

Languages: Python3, C++, C, Go  
github.com/MakArtKar  
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leetcode.com/MakArtKar

## SKILLS

**Libraries** PyTorch, JAX, Transformers, Hydra, DVC, WandB, OpenCV, Sklearn, Numpy, Pandas, ONNX, Seaborn  
**Tools** SLURM (DDP, FSDP), SQL, Linux, Bash, Git, LaTeX, Markdown

## TECHNICAL EXPERIENCE

 **JetBrains Research** Feb 2025 — Present  
ML Researcher with D. Vetrov Python 3, PyTorch, JAX, transformers, Hydra, Lightning, SLURM

- Latent Diffusion Drafters for speculative decoding: lossless acceleration of LLM inference
- Distillation (flow maps, consistency) and latent space compression for real-time acceleration
- Masked Diffusion LMs
- Inference-time scaling, diffusion drafters for speculative decoding; large-scale models (Dream-7B, Qwen3-0.6B MDLM/BD3LM)
  - **G-Star**: ICML 2026 main track; multi-node training; PyTorch, JAX

 **Yandex** Jun 2023 — Sep 2024  
Middle ML Engineer/ Geo services Python 3, LangChain, PyTorch, Hydra, OpenCV, YT, Shapely, YQL, Nirvana

- Map roads segmentation from aerophotos with IoU 0.93
- Roads post-processing algorithms - reduced number of artifacts from 10% → 0.02% and improved visual quality.
  - Safety islands segmentation and vectorization with IoU 0.37
- News classification to create map updates hypotheses. Precision: 0.9, recall: 0.85.
- Developing and maintenance of the news pipeline: site parsing (with LangChain AI agents), structuring data and classifying.

 **Sber Devices** Jul 2022 — Jun 2023  
Middle ML Engineer/ Computer Vision Platform Python 3, C++, PyTorch, Hydra, DVC, OpenCV, ONNX, torch\_pruning

- Background removal for video conferencing with 0.98 dice metric
- Improved human segmentation quality working on segmentation pipeline, pruned model getting 1.5x speed acceleration
  - Improved visual quality with post-processing algorithms: aligning the lighting and smoothing out the border.
- Trained 700KB model for pay card border segmentation and developed C++ inference pipeline in multiplatform mobile SDK

 **AIRI (Artificial Intelligence Research Institute)** Jul 2022 — Oct 2022  
ML Research Intern / GAN adaptation Python 3, PyTorch, Hydra, WandB

- **Few Shot Image Generation** implementation for GAN domain adaptation to a new domain from ~10 images

 **Yandex** Jul 2021 — Oct 2021  
ML Developer intern / VR Team with V. Lempitsky Python 3, PyTorch, Tensorboard

- Trained model that predicts human's alpha mask to improve the quality of a VR scene
- Prepared datasets with human alpha masks, read and compared papers on an alpha matting problem
  - Trained an improved **FBA Matting** net without using trimaps

 **Samsung R&D Department** Nov 2020 — June 2021  
ML Research Intern / Mobile Authentication Python 3, PyTorch, Bash, Numpy, Pandas, Sklearn, Tslearn

- Worked on a pipeline for mobile phone authentication by user's motion patterns with  $82.2\% \pm 9\%$  accuracy - [paper](#).
- Collected, preprocessed and handled data from devices. Used ML and DL models for authentication, got 70% accuracy baseline

## EDUCATION

 **PhD in Computer Science (1st year)**, Constructor University of Bremen, Germany Sep 2026 — Present

 **Master in Computer Science**, Constructor University of Bremen, Germany Sep 2024 — May 2026

 **Bachelor in Computer Science**, GPA 86/100, Neapolis University of Paphos, Cyprus Jan 2023 — May 2024

 **Bachelor in Computer Science**, GPA 8.7/10, Higher School of Economics, Moscow, Russia Sep 2019 — Jun 2024

▪▪ **Master's Equivalent**, Yandex School of Data Analysis, Data Science, Russia Sep 2020 — May 2022

Courses: Deep Vision, CV, NLP, 3D CV, Generative Models, RL, Bayes Methods, Optimization Methods, Deep Bayes, Large Scale DL

Courses: ML, DL, Python, C++, Go, Algorithms and Data Structures, Matrix Calculations, Statistics, Linear algebra, Probability theory

## PROJECTS

**Telegram RAG QA bot** Jun 2023 — Aug 2023  
→ Developed tg bot to answer on questions giving it the chat context Python 3, LangChain, HuggingFace, aiogram

- Researched different LLM models, developed different types of LangChain pipeline (stuff, rerank, map reduce, refine) with them.

**Automatic Detection of Means of Transportation on 3d Cargo Models** Dec 2021 — June 2022  
→ Developed model for cloud segmentation with 0.95 IoU for cargo Python 3, Pytorch, PyTorch3D

- Calibrated cameras for cargo scanning, prepared data, implemented ESANet model